

CSCE 645 GEOMETRIC MODELING · MODULE 09 · LECTURE NOTES

Parameterization

Flattening a surface, and choosing which distortion to accept.

MODULE OUTCOMES

- Explain why distortion is unavoidable and quantify it.
- Distinguish conformal, authalic, and isometric maps.
- Implement harmonic and least-squares conformal parameterisation.
- Explain cutting, charts, and atlas packing.
- Diagnose a bad UV map from its distortion measure.

1 Distortion is unavoidable

Parameterisation assigns a 2D coordinate to every point of a surface — the UV map of CSCE 641 Module 08, now constructed rather than assumed. The first thing to establish is that it cannot be done perfectly.

THE ARGUMENT, IN THREE LINES

Gaussian curvature is **intrinsic** (Module 07): computable from lengths and angles measured within the surface, with no reference to the ambient space.

An **isometric** map preserves all lengths, hence all intrinsic quantities, hence K .

A sphere has $K = 1/r^2 \neq 0$; a plane has $K = 0$. Therefore no isometric map between them exists.

This is a theorem. No future algorithm will flatten a sphere without distortion. The entire subject is about *choosing which distortion to accept*.

Map type	Preserves	Distorts	When it exists
Isometric	Lengths, angles, areas — everything.	Nothing.	Only for developable surfaces: planes, cylinders, cones. K must already be 0.
Conformal	Angles. Locally, shapes are preserved and only scaled.	Area — regions grow or shrink.	Any disc-topology patch. Guaranteed by the uniformisation theorem.
Authalic	Area.	Angles — shapes shear.	Any disc-topology patch.
Neither	—	Both, but each by less.	Always. ARAP and similar methods live here.

THIS IS THE MAP PROJECTION PROBLEM

Mercator is conformal: angles and local shapes are right, which is why it was used for navigation, and areas are badly wrong — Greenland appears the size of Africa.

Equal-area projections get the areas right and shear the shapes.

Cartographers have argued about this for four centuries and the constraint has never moved, because it is Theorema Egregium. UV unwrapping is the same problem with a different surface.

1.1 Which distortion to accept

For texture mapping the answer is nearly always conformal, and the reason is perceptual rather than mathematical.

- **Angle distortion** appears as shearing: a checkerboard becomes a pattern of parallelograms, straight features bend, text becomes italic. The eye detects this immediately.
- **Area distortion** appears as varying texel density: some regions are sharper than others. The eye tolerates this well unless the variation is extreme.
- So: preserve angles, and control area distortion separately by cutting the surface into charts small enough that the area variation within each is acceptable.

2 Harmonic parameterisation

The simplest method, and it is a direct application of Module 08.

- 1 Cut the surface to disc topology (§4).
- 2 Map the boundary to a convex shape — a circle or square — by arc length.
- 3 Solve $\Delta u = 0$ and $\Delta v = 0$ in the interior, with those boundary values fixed.

That is two sparse linear solves with the Laplacian you already built, with boundary rows replaced by identity constraints. The resulting map is harmonic: the smoothest possible interpolation of the boundary conditions.

TUTTE'S EMBEDDING THEOREM

If the boundary is mapped to a **convex** polygon and all edge weights are **positive**, the resulting map is guaranteed to be **bijective** — no triangle flips, no fold-over.

This is an unusually strong guarantee and it is why harmonic parameterisation remains worth knowing despite its distortion.

But the positivity condition matters. Cotangent weights are negative on obtuse triangles (Module 07), which violates the hypothesis and allows flips. Using uniform weights restores the guarantee at the cost of ignoring geometry; remeshing to remove obtuse triangles (Module 10) keeps both. This is a concrete case where mesh quality determines whether a theorem applies.

The real limitation is the fixed boundary. Forcing a long thin patch onto a circle stretches it severely, and the distortion is imposed by your boundary choice rather than by the surface. That is what free-boundary methods fix.

3 Free-boundary methods

3.1 Least-squares conformal maps

A map is conformal exactly when it satisfies the Cauchy–Riemann equations. LSCM does not attempt to satisfy them exactly — which is generally impossible on a triangulation — but minimises the squared violation over the whole mesh.

Because no boundary is prescribed, the system has a four-dimensional null space corresponding to translation, rotation, and uniform scale. Pinning **two** vertices removes it. The result is a sparse least-squares problem, solved once, and the boundary takes whatever shape minimises distortion.

Distortion is typically far lower than harmonic parameterisation with a forced boundary, and LSCM or a close relative is what production UV tools run when you press 'unwrap'.

Method	Boundary	Guarantee	Distortion	Cost
Harmonic / Tutte	Fixed, convex	Bijjective (positive weights)	High when the boundary is forced	Two sparse solves
LSCM	Free, two pins	Usually bijective in practice	Low	One least-squares solve
ABF++	Free	Near-conformal; angle-based formulation	Very low	Nonlinear, iterative
ARAP	Free	—	Low; balances angle and area	Local/global iteration
BFF	Free	Conformal, with boundary control	Very low	Fast; essentially two solves

Boundary First Flattening (Sawhney and Crane) is worth knowing about: it is conformal, extremely fast, and gives direct control over the boundary shape or the boundary scale factor, which is exactly what an artist wants.

4 Cutting and atlases

Every method above requires **disc topology**. A sphere, a torus, and a character are not discs, so the surface must be cut first.

4.1 Seams

Cutting introduces **seams**: curves along which the texture coordinates are discontinuous. Across a seam, filtering and mipmapping break down (CSCE 641 Module 08), and any mismatch in the texture shows as a visible line.

THE THREE-WAY TRADE, AGAIN

CSCE 641 Module 08 claimed that every UV map trades **stretch**, **seams**, and **wasted atlas space** against each other, and that none can be eliminated. This module is why.

More cuts → each chart is flatter → less distortion — but more visible seams and more wasted space between packed charts.

Fewer cuts → cleaner appearance and better packing — but higher distortion within each chart.

There is no setting that wins on all three, and the balance depends on the asset. That is why UV layout remains skilled manual work on important models.

Where to cut: in regions of high Gaussian curvature, since that is where the most distortion is relieved per unit of seam length; along existing feature lines and material boundaries, which are already discontinuities; and in places that will not be looked at. Automatic seam generation handles the first two reasonably and cannot evaluate the third, which is why artists still place seams by hand on hero assets.

4.2 Measuring distortion

Per triangle, the map has a 2×2 Jacobian with singular values $\sigma_1 \geq \sigma_2$ — the stretch factors along two principal directions.

Quantity	Meaning	Ideal
σ_1/σ_2	Anisotropy — angle distortion	1 (conformal)
$\sigma_1\sigma_2$	Area scaling	1 (authalic)
$\max(\sigma_1, 1/\sigma_2)$	Worst-case stretch	1
sign of $\det J$	Orientation	Positive everywhere, or a triangle has flipped

VISUALISE IT

Colour each triangle by σ_1/σ_2 and the failures of a UV map become obvious at a glance: stretched regions light up, and flipped triangles show as negative determinant.

This is the most useful debugging tool in the module and is required in Project 2. A checkerboard texture applied to the model is the artist's version of the same diagnostic, and it works for the same reason.

4.3 Packing

Charts must be arranged into one rectangular texture. This is 2D bin packing, which is NP-hard (CSCE 629 Module 12), so every implementation is a heuristic.

- **Gutters.** Leave empty space between charts so that mipmap filtering does not blend one chart into its neighbour. The gutter must be wide enough for the coarsest mip level you intend to use — which means it scales with the mip count, not with a fixed pixel value.

- **Consistent texel density.** Charts should be scaled so that equal surface area receives equal texture area, or some parts of the model will be visibly sharper than others.
- **Texel alignment.** Aligning chart boundaries to texel centres avoids resampling artifacts at the edges.
- **Utilisation.** 70–85% is typical and considered good. The remainder is gutters and packing waste.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Keenan Crane — DDG, conformal geometry lectures	Conformal maps, the uniformisation theorem, and why angles are the right thing to preserve.
Lévy et al. — Least Squares Conformal Maps (free)	The LSCM paper. Short, clear, and directly implementable.
Sawhney & Crane — Boundary First Flattening (free)	A fast conformal method with direct boundary control, plus a free tool to compare your results against.
Hormann, Lévy & Sheffer — Mesh Parameterization SIGGRAPH course (free)	The comprehensive survey. The reference when you need a method this module did not cover.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement harmonic parameterisation: map the boundary to a circle and solve $\Delta u = 0$ twice. Apply a checkerboard texture and capture the distortion.
- 2 Repeat with uniform weights and with cotangent weights on a mesh containing obtuse triangles. Find a flipped triangle in the cotangent version and explain it via Tutte's hypothesis.
- 3 Implement LSCM with two pinned vertices. Compare distortion against the harmonic result on the same patch, reporting the distribution of σ_1/σ_2 .
- 4 Implement per-triangle distortion visualisation. Colour by anisotropy and by area scaling, and render both.
- 5 Cut a closed mesh into charts by hand, parameterise each, and pack them into an atlas. Report the utilisation percentage.
- 6 Experiment with seam placement: parameterise the same model with two seams and with eight. Compare maximum distortion and atlas utilisation, and render both with a checkerboard.
- 7 Implement gutter generation and demonstrate texture bleeding across charts at a coarse mip level when the gutter is too narrow.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why is distortion-free flattening impossible? Give the three-line argument.
- 2 What do conformal and authalic maps each preserve, and why does texture mapping prefer conformal?
- 3 Describe harmonic parameterisation in three steps, and state Tutte's guarantee with its hypotheses.
- 4 Why can cotangent weights break Tutte's guarantee, and what are your options?
- 5 What does LSCM minimise, and why must two vertices be pinned?
- 6 What are the three quantities a UV layout trades against each other?
- 7 How do you measure distortion per triangle, and what would a flipped triangle look like in that measure?